

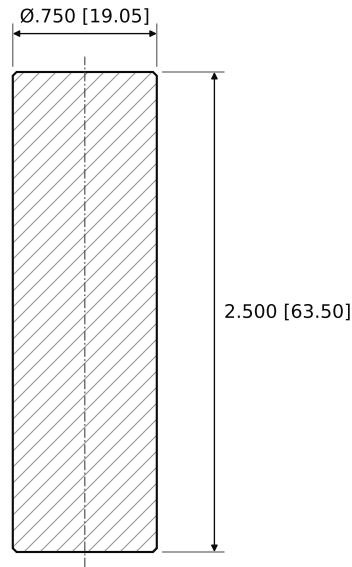
rePowder first runs: parts for the prototyping lab

Every part is shown cut through the middle; hatching is metal. Make one P1 first and check it gets past the sealing rod's adapter into the crucible before making cups (README) · issue #222

P1 Solid slug

6063-T52 · qty 2 (E2)

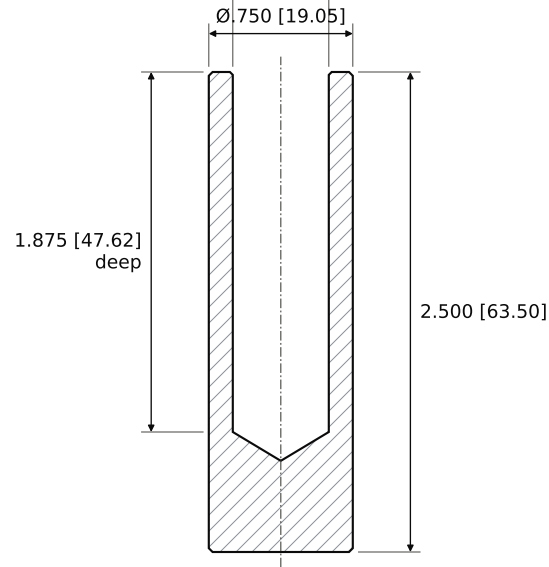
OD as received · $.02 \times 45^\circ$ both ends



P2 Cup

6063-T52 · qty 10 (E1, E3-E6)

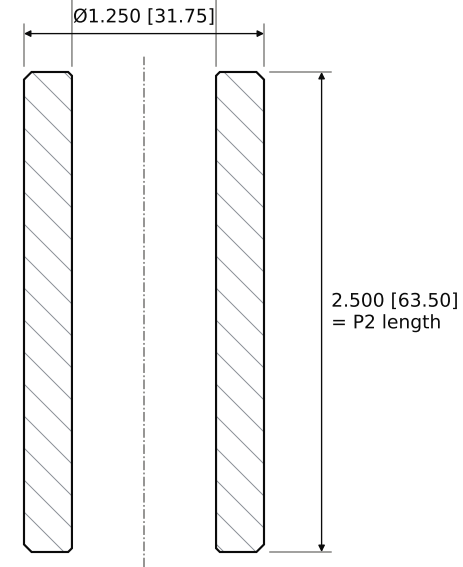
drill 31/64, bore or ream · 118° bottom is fine
 $\varnothing .500$ [12.70] bored



F1 Press sleeve

1018 cold-finished (A108) · qty 1 · only for E4

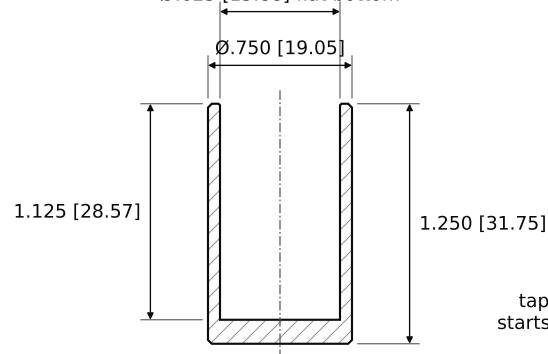
stops the cup wall bulging under the press
 $\varnothing = \text{measured P2 OD} + .0005 \text{ max}$



P4 Small cup

6063-T52 · qty 2 (E7, later)

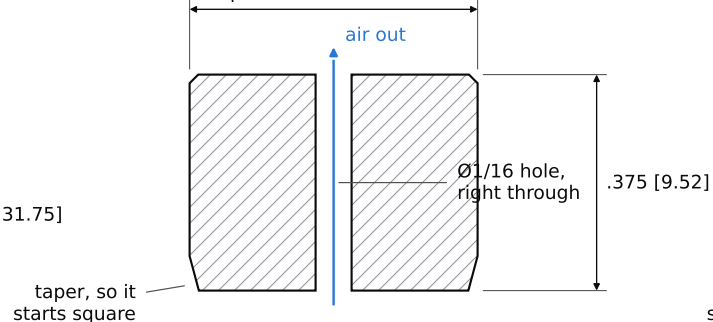
1/16" wall · $.125$ [3.17] floor
 $\varnothing .625$ [15.88] flat bottom



P3 Lid for P2 (3:1)

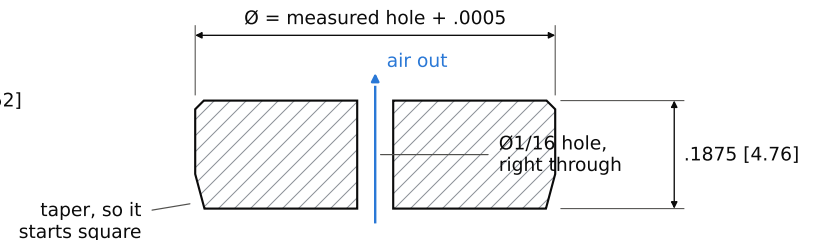
6063-T52 · qty 10 · one per cup, not interchangeable

$\varnothing = \text{that cup's measured hole} + .0005-.0008$



P5 Lid for P4 (3:1)

6063-T52 · qty 2 · same hole and taper as P3



NOTES

- 1 All parts from the 3/4" 6063-T52 bar on hand (McMaster 1640T16). Inches [mm].
- 2 Break the outside edges $.020 \times 45^\circ$ and deburr the holes. No marker, no stamping.
- 3 Make a cup, measure its hole, then turn that cup's lid $.0005-.0008$ " bigger, so it presses in. No more than $.0008$ " : at $.001$ " the bore is past yield (von Mises 123 MPa). Bag each lid with the cup it was made for - they are not interchangeable.

- 4 Every lid gets the $\varnothing 1/16$ hole. The chamber is pumped down before melting, and air shut under a solid lid has to escape through the powder instead (see #104).
 - 5 E4 only: stand the filled cup inside F1 and press the lid flush. F1 is the same length as the cup, so the press bottoms out at flush. Push the cup out with a 5/8" drift.
 - 6 Degrease in IPA, dry, weigh each part to 0.01 g, then bag and label it.
- Parts 1:1 on Letter at 100 %, lids 3:1 · STEP files: atomizer-charge/cad/step/